AI INTERACTION LOG – Sprint 2

Title: Implement Job Application Submission and Status Tracking API

Purpose of AI Use: Database migration design, Express.js route implementation, and unit test debugging.

Prompt/Response: Prompts used: "Help design a PostgreSQL migration script for a job applications table in an Express.js backend that enforces valid statuses (Applied, Under Review, Interview, Offered, Rejected) and prevents duplicate applications per user", plus debugging prompts for pg-mem test runner integration.

AI-Suggested Content: Provided the SQL schema structure using SERIAL PRIMARY KEY, foreign keys, CHECK constraints on valid status strings, Express routes (POST /api/applications, GET /api/applications/me, PATCH /api/applications/:id/status), and automated test cases using Node's native test runner (node --test).

Validation: Tested the implementation locally using npm test. Confirmed that the test suite runs successfully with pass 1 and fail 0, validating schema migrations, constraints, status filters, and duplicate application handling.

Reflection: The AI assistance was highly efficient for structuring relational database constraints correctly and establishing robust Express routing and endpoint testing without manual documentation lookup.

Responsible Person: Jack Wu